

Prevention and Management of Congenital Syphilis

Prevention of Congenital Syphilis

- **Recommendation:** All pregnant women should be screened for syphilis at the first antenatal visit (first trimester).
- Serological testing should be repeated in the third trimester and at delivery in areas of high prevalence or for women at increased risk.
- Some specialists recommend that all infants born to mothers with positive syphilis serology receive a single dose of **benzathine benzylpenicillin (BPG) 50,000 units/kg IM**, regardless of whether the mother was treated during pregnancy.

Diagnosis of Congenital Syphilis

Confirmed Congenital Infection

- *Treponema pallidum* identified by dark-field examination (DFE) or PCR in:
 - Placenta or autopsy material
 - Exudate from skin/mucosal lesions
 - Body fluids (e.g., nasal discharge)

Presumed Congenital Infection

- A stillborn infant with a positive treponemal test.
- Live-born infant with a positive treponemal test **and** one or more of the following:

Clinical Signs:

- Persistent rhinitis, condylomata lata
- Osteitis, periostitis, osteochondritis
- Ascites, cutaneous or mucous membrane lesions
- Hepatitis, hepatosplenomegaly

- Glomerulonephritis, haemolytic anaemia

Radiological Findings:

- Abnormalities of long bones on X-ray suggestive of congenital syphilis

Laboratory/Cerebrospinal Fluid (CSF) Findings:

- Positive RPR/VDRL in CSF
- Fourfold or greater increase in TPPA/TPHA titre in infant compared to mother (samples taken simultaneously at birth)
- Fourfold or greater increase in non-treponemal test titre in infant compared to mother (samples taken simultaneously at birth)
- Fourfold or greater rise in non-treponemal test titre within 3 months after birth
- Positive anti-*T. pallidum* IgM assay (e.g., IgM EIA, 19S-IgM FTA-abs, or IgM immunoblot) in infant's serum

Maternal History:

- Mother diagnosed with syphilis during pregnancy but not adequately treated before or during pregnancy

Late Presentation:

- Child >12 months of age with positive treponemal serology and sexual abuse excluded

Late Congenital Syphilis

Clinical manifestations include: - Interstitial keratitis - Clutton's joints - Hutchinson's incisors, mulberry molars - High palatal arch - Rhagades - Sensorineural deafness - Frontal bossing, short maxilla, mandibular protuberance - Saddlenose deformity - Sternoclavicular thickening - Paroxysmal cold haemoglobinuria - Neurosyphilis or gummatous lesions

Note: Serological tests may be negative in infants infected late in pregnancy; repeat testing is recommended.

Even if the mother is treated in the third trimester, treatment may be insufficient to prevent congenital syphilis in the infant.

Reporting: All cases must be reported to the national Syphilis Surveillance system where mandated.

Investigations

- **Serology** (on infant's blood, **not umbilical cord blood**):
 - RPR/VDRL (quantitative)
 - TPPA/TPHA (quantitative)
 - Anti-treponemal IgM assays (IgM EIA, 19S-IgM FTA-abs, IgM immunoblot)
- **Blood tests:**
 - Full blood count
 - Liver function tests
 - Electrolytes
- **CSF analysis:**
 - Cell count, protein, RPR/VDRL, TPHA/TPPA
- **Imaging:**
 - X-rays of long bones
- **Ophthalmic assessment** as clinically indicated

First-Line Therapy

For confirmed or presumed congenital syphilis:

- **Benzympenicillin 150,000 units/kg/day IV** in six divided doses every 4 hours for 10–14 days [IV; C]

If CSF is normal:

- **Option a (First-line):**

Benzathine benzylpenicillin (BPG) 50,000 units/kg IM (single dose), up to adult dose of 2.4 million units [IV; C]

- **Option b (Second-line, if BPG is not available):**

Procaine penicillin 50,000 units/kg/day IM for 10–14 days [IV; C]

HIV-Infected Patients

- Serological tests for syphilis are generally reliable in HIV-positive individuals for both diagnosis and monitoring treatment response.
- Close follow-up is recommended due to potential atypical serological responses and higher risk of treatment failure or progression to neurosyphilis.